

Benchmark Investigation of Panda: A Pretrained Forecast Model for Chaotic Dynamics

Experiment Log and Analysis

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Abstract

This document records all benchmark experiments conducted on Panda (Lai, Bao, Gilpin, ICLR 2026), a pretrained forecast model for chaotic dynamical systems. The investigation was motivated by Prof. Salim’s direction to test Panda’s emergent capabilities on standard time series and spatiotemporal benchmarks. Each experiment is reported with its motivation, methodology, methodological limitations, direct observations, and competing explanations. Claims are explicitly labelled as: **[OBS]** observed in data, **[PAT]** empirical pattern across multiple conditions, **[HYP]** proposed explanation, **[SPEC]** speculative, or **[EST]** established by prior theory or paper. This labelling convention was decided to maintain distinction between observation and possible interpretations rather than mixing them.

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1 Experimental Setup

1.1 Models

Two models are compared throughout all experiments.

Panda 21M. The base checkpoint from `GilpinLab/panda` on HuggingFace. A 21 million parameter encoder-only transformer trained exclusively on 20,000 synthetic chaotic ordinary differential equations (ODEs). Prediction horizon 128 steps. Context length 512 steps. For horizons exceeding 128 steps, window autoregression is used: predictions are fed back as context for subsequent rollouts.

Chronos 20M. The `amazon/chronos-t5-small` checkpoint. A 20 million parameter causal univariate decoder model trained on a large corpus of real-world time series. Chronos forecasts each channel independently.

The two models are matched in parameter count, making comparisons attributable to architecture and training distribution rather than scale.

1.2 Evaluation Protocol

Unless stated otherwise, the following protocol is used across all experiments.

Normalisation. Each context window is normalised independently using its own mean and standard deviation per channel (per-window instance normalisation). This prevents global statistics from leaking future information into each evaluation window.

Evaluation windows. A sliding window of context length 512 is moved across the time series. The number of windows is specified per experiment. In early experiments (pre-presentation), $n_{\text{windows}} = 6\text{--}10$ was used. In the fixed experiment set, $n_{\text{windows}} = 8$ or $n_{\text{windows}} = 20$ is used as noted.

Primary metric. Mean Absolute Error (MAE) is the primary metric throughout. Symmetric Mean Absolute Percentage Error (sMAPE) was used in earlier experiments but was found to be unreliable on instance-normalised data where values near zero cause the denominator to vanish, artificially inflating or deflating the metric.

Statistical test. Where stated, the Wilcoxon signed-rank test (one-sided, alternative: model A better than model B) is applied to the per-window MAE values. With $n_{\text{windows}} = 8$, the minimum achievable p-value is 0.008. With $n_{\text{windows}} = 20$, the minimum achievable p-value is approximately 0.0001. All p-values should be interpreted in light of these floors.

Advantage. Panda advantage is defined as $\text{MAE}_{\text{Chronos}} - \text{MAE}_{\text{Panda}}$. Positive values indicate Panda is better.

1.3 Hardware

All experiments were run on a CPU-only laptop. Runtimes are consequently long for experiments involving many channels or long horizons.

2 Pre-Presentation Experiments

These experiments were conducted prior to the group meeting and form the basis of the initial findings presented to Prof. Salim.

2.1 Experiment 1: Standard Time Series Benchmarks (Initial)

2.1.1 Motivation

To establish whether Panda has any advantage on non-chaotic standard benchmarks. ETTh1, ETTh2, and Weather were chosen as the canonical time series forecasting benchmarks used by TimesNet, PatchTST, and Chronos comparisons in the literature.

2.1.2 Method

Datasets: ETTh1 (7 channels, 17,420 timesteps), ETTh2 (7 channels, 17,420 timesteps), Weather (21 channels, 52,696 timesteps). Prediction horizon: 128 steps. $n_{\text{windows}} = 10$. Primary metric: sMAPE. Instance normalisation applied globally to the full dataset before windowing (not per-window).

2.1.3 Methodological Limitations

1. sMAPE is unreliable on instance-normalised data where channel values are near zero. Results are not quantitatively trustworthy.
2. Global normalisation leaks future statistics into evaluation windows.
3. Single prediction horizon (128 steps). Results may not generalise to other horizons.
4. No statistical significance tests.
5. $n_{\text{windows}} = 10$ provides limited statistical power.

2.1.4 Observations

[OBS] Both models produced sMAPE approximately 0.8–0.9 on all three datasets. **[OBS]** Panda slightly worse than Chronos on ETTh1. **[OBS]** Chronos sMAPE = 0.009 on Lorenz rho = 15 was flagged as a metric artifact rather than genuine model performance.

2.1.5 Explanation Shipped at the Time

The conclusion presented was: “Panda fails on non-chaotic data because it was trained on stationary chaotic ODEs with no seasonal inductive bias.” This was stated as a finding despite the invalid metric and the limitation to a single prediction horizon.

2.1.6 Alternative Explanations Not Presented

1. **[HYP]** The result may be specific to $H = 128$. Later fixed experiments showed Panda wins on Weather at all tested horizons and on ETTh1 and ETTh2 at longer horizons.
2. **[HYP]** The sMAPE artifact may have masked the true comparison entirely, making the result uninterpretable.
3. **[HYP]** Weather and ETT differ substantially in channel count and coupling structure. Grouping them as “non-chaotic data” ignored this distinction.

Confidence: Very Low. The original conclusion is largely contradicted by the fixed Experiment 8.

2.2 Experiment 2: Double Pendulum Graded Noise

2.2.1 Motivation

To quantify the cost of measurement noise on a chaotic system and test whether Panda’s advantage over Chronos is robust to noise. This also served as a proxy for the gap between clean synthetic data and noisy real experimental data.

2.2.2 Method

The double pendulum ODE was simulated with Gaussian observation noise at six levels: $\sigma \in \{0, 0.01, 0.05, 0.10, 0.25, 0.50\}$ times the signal standard deviation. One random seed, one initial condition. $n_{\text{windows}} = 8$. sMAPE and MAE both reported.

2.2.3 Methodological Limitations

1. Single seed and single initial condition. One realisation of the trajectory.
2. The real experimental dataset (Asseman et al. 2018) was not obtained. The experiment was described as “synthetic versus real” but was in practice “synthetic versus synthetic with Gaussian noise.” Gaussian additive noise is not the same type of nonstationarity as real experimental measurement error.
3. Noise levels were not tested beyond $\sigma = 0.50$.
4. No statistical significance tests.

2.2.4 Observations

[OBS] Panda MAE at $\sigma = 0$: 0.62. Panda MAE stays approximately flat (0.63–0.67) across all tested noise levels up to $\sigma = 0.50$. **[OBS]** Chronos MAE at $\sigma = 0$: 0.71. Chronos MAE peaks at approximately 0.77 at high noise. **[OBS]** Panda consistently lower MAE than Chronos at every tested noise level. **[OBS]** sMAPE panel showed erratic non-monotone behaviour, correctly identified as a metric artifact on normalised data.

2.2.5 Explanation Shipped at the Time

“The random Fourier and polynomial features are smooth functions of the input, making the embedding noise-robust. Takens’ theorem is noise-tolerant.” This was stated as a conclusion rather than a hypothesis.

2.2.6 Alternative Explanations Not Presented

1. **[HYP]** Chronos uses a discrete tokenisation scheme. Continuous Gaussian noise may specifically disrupt this tokenisation, independently of any property of Panda’s architecture.
2. **[HYP]** Panda’s predictions may have lower variance by default. A flat MAE noise curve could reflect low-variance predictions rather than genuine noise robustness.
3. **[HYP]** The tested noise levels may be below the breakdown threshold for both models. The claim cannot be extrapolated to higher noise regimes.

Confidence: Medium. for the observation. **Confidence: Low.** for the proposed mechanism.

2.3 Experiment 3: Lorenz Rho Sweep

2.3.1 Motivation

To test whether Panda’s advantage specifically tracks the onset of chaos. The Lorenz parameter ρ controls the transition from periodic ($\rho < 24.74$) to chaotic ($\rho \geq 24.74$) dynamics. This provides a controlled test of the chaos-threshold hypothesis.

2.3.2 Method

$\rho \in \{10, 15, 20, 24, 24.74, 26, 28, 35, 45, 60\}$. One trajectory per ρ value, one initial condition. $n_{\text{windows}} = 8$. MAE as primary metric. Maximum Lyapunov exponent estimated using the Rosenstein method. The estimator required three implementation attempts before producing physically reasonable output.

2.3.3 Methodological Limitations

1. The Lyapunov estimator systematically overestimates. For the Lorenz system at $\rho = 28$, the literature value is $\lambda_1 \approx 0.9$ but the estimator returned 1.66. The x-axis of Lyapunov-based figures is not quantitatively reliable.
2. Single trajectory and single seed per ρ value.
3. $n_{\text{windows}} = 8$ with no confidence intervals on advantage values.
4. Results at $\rho = 10, 15$ showed Chronos sMAPE near zero in the periodic regime, correctly identified as an artifact.

2.3.4 Observations

ρ	Regime	$\hat{\lambda}_1$	Panda MAE	Chronos MAE	Advantage
10.00	Periodic	-0.29	≈ 0.02	≈ 0.02	≈ 0
15.00	Periodic	-0.33	≈ 0.04	≈ 0.04	≈ 0
20.00	Near transition	0.20	≈ 0.09	≈ 0.31	+0.22
24.00	Near transition	0.33	≈ 0.10	≈ 0.44	+0.34
24.74	Onset of chaos	1.75	≈ 0.18	≈ 0.65	+0.47
28.00	Chaotic	1.66	≈ 0.25	≈ 0.89	+0.64
60.00	Chaotic	1.63	≈ 0.48	≈ 1.17	+0.69

Table 1: Lorenz ρ sweep results (approximate values from $n_{\text{windows}} = 8$, one seed).

[OBS] Estimated λ_1 is negative at $\rho = 10, 15$ and jumps to 1.75 at $\rho = 24.74$, consistent with the theoretical bifurcation point. **[OBS]** Panda MAE advantage is approximately zero at $\rho = 10, 15, 20$. The advantage jumps near $\rho = 24.74$ and remains positive across all tested chaotic ρ values. **[OBS]** No tested chaotic ρ value produced a result where Chronos had lower MAE than Panda. **[PAT]** The pattern of advantage increasing with ρ is consistent across the chaotic range, though not strictly monotone.

2.3.5 *Explanation Shipped at the Time*

“Panda’s advantage is specifically activated by the chaotic transition. The positive Lyapunov exponent is necessary and sufficient for Panda’s advantage.” This was stated as a conclusion.

2.3.6 *Alternative Explanations Not Presented*

1. **[HYP]** Signal statistics (variance, spectral content, amplitude) change discontinuously at the bifurcation. Any model better suited to high-variance aperiodic signals would show this pattern, independently of any chaos-specific mechanism. This alternative is not yet ruled out.
2. **[HYP]** Chronos tokenisation may fail specifically on aperiodic continuous signals, making Chronos worse in the chaotic regime regardless of Panda’s capabilities.
3. **[HYP]** The advantage may be driven primarily by the Koopman dynamics embedding rather than any chaos-specific learning. This is not testable without an ablation removing the dynamics embedding.

What would falsify the claim: A chaotic system where Panda does not win, or a non-chaotic system where Panda wins substantially. The Weather and Burgers sweep results (Experiments 8 and 10) partially address this.

Confidence: Medium. for the observation. **Confidence: Low.** for the chaos-specific mechanism hypothesis.

2.4 Experiment 4: `dysts` Systems and Advantage versus `Lambda1`

2.4.1 *Motivation*

To test Panda’s advantage across a broader set of known chaotic systems from the `dysts` library and to measure whether a Spearman correlation between estimated λ_1 and Panda advantage reaches statistical significance.

2.4.2 *Method*

Systems tested: Lorenz, Rossler, Chua, Duffing, Halvorsen, SprottB, Thomas, Rucklidge, Dadras, Bouali, DequanLi. $n_{\text{windows}} = 8$ per system. Lyapunov estimates via Rosenstein method. Additional systems were added iteratively until the Spearman p-value crossed 0.05.

2.4.3 *Methodological Limitations*

1. **P-hacking.** Systems were added iteratively until $p < 0.05$ was achieved. The reported Spearman $\rho = 0.370$, $p = 0.044$ is not a valid pre-registered result. The true p-value corrected for sequential testing is unknown but higher.
2. Systems in the `dysts` library are not independent (several share structural similarities).
3. Lyapunov estimates are noisy and systematically biased (see Experiment 3).
4. Single seed per system. No cross-system multiple comparison correction.

2.4.4 Observations

[OBS] Panda wins (lower MAE) on every tested `dysts` chaotic system. [OBS] Spearman $\rho = 0.370$, $p = 0.044$ after iterative addition of systems. This p-value is not reliable due to p-hacking. [OBS] Two visual sub-clusters: Lorenz family above the trend line, double pendulum below the trend line.

2.4.5 Explanation Shipped at the Time

“Panda has learned something universal about chaotic dynamics. Chaos is necessary and sufficient. In-distribution systems show larger advantage.” Overstated given the methodological problems.

2.4.6 Alternative Explanations Not Presented

1. [HYP] The correlation may be driven entirely by the Lorenz family cluster. Removing those points may eliminate significance entirely.
2. [HYP] The sub-cluster structure (Lorenz family above trend, double pendulum below) suggests training-distribution effects rather than universality. This directly challenges the “universal chaos learning” claim.

Confidence: Low. due to p-hacking. The observation that Panda wins on every tested chaotic system is more robust (**Confidence: Medium.**) but still limited to $n = 8$ windows per system.

2.5 Experiment 5: Burgers Viscosity Sweep (First Version)

2.5.1 Motivation

To test whether the chaos-threshold pattern found in the Lorenz rho sweep extends to partial differential equations (PDEs). Burgers equation viscosity ν controls the transition from smooth diffusion-dominated to chaotic shock-dominated dynamics, analogous to ρ in Lorenz.

2.5.2 Method

$\nu \in \{0.1, 0.05, 0.02, 0.01, 0.005\}$. PCA reduction to 16 channels. $n_{\text{windows}} = 8$. Lyapunov estimated from first PCA component. Solver diverged at $\nu = 1.0$, leaving no non-chaotic baseline.

2.5.3 Methodological Limitations

1. No non-chaotic baseline. Without $\nu \geq 0.5$, it is impossible to verify a threshold effect.
2. Lyapunov estimation returned approximately zero for all conditions. The chaos-threshold connection could not be verified from the data.
3. Non-monotone reversal of advantage at $\nu = 0.02 \rightarrow 0.01$ was not discussed.
4. PCA representation changes character as ν changes, confounding the comparison across conditions.

2.5.4 Observations

[OBS] Panda advantage positive at all tested ν values. **[OBS]** Advantage roughly increases with decreasing ν but with a non-monotone reversal at $\nu = 0.02 \rightarrow 0.01$.

Current status: Superseded by the fixed Experiment 10, which includes a non-chaotic baseline and a stable solver.

2.6 Experiment 6: PCA versus Spatial Subsampling (First Version)

2.6.1 Motivation

To compare two methods for converting a 1D spatiotemporal PDE field into channels for Panda: global PCA modes versus raw spatial location subsampling.

2.6.2 Method

Single ν value (0.005). Single spatial location ($x = 0$) for subsampling. $n_{\text{windows}} = 6$.

2.6.3 Critical Flaw

The spatial location $x = 0$ was a near-nodal point of the Burgers field with near-constant dynamics. Both models trivially predicted near-zero values and scored well. No general conclusion about spatial subsampling is possible from one degenerate location. This experiment was rerun in fixed Experiment 12 with four methods including a variance-stratified control.

2.6.4 Observations

[OBS] Both models performed approximately $8\times$ better in absolute MAE on spatial subsampling than on PCA. **[OBS]** Panda relative advantage larger with PCA (0.093) than subsampling (0.044).

Confidence: Very Low. for any conclusion. Single degenerate location.

3 Fixed Experiments

These experiments address the methodological problems identified in the pre-presentation set. A new notebook (`fixed_experiments.ipynb`) was created. Key improvements: per-window normalisation, increased n_{windows} , Wilcoxon signed-rank tests, multiple prediction horizons, non-chaotic baselines, and removal of oracle leakage.

3.1 Experiment 7: Standard Horizon Evaluation (Week 1, Invalid)

The Week 1 version of standard horizon evaluation retained global normalisation leakage and used $n_{\text{windows}} = 6$ with sMAPE as primary metric. Results were superseded by Experiment 8 and are not reported here.

3.2 Experiment 8: Standard Horizon Evaluation (Fixed)

3.2.1 Motivation

To establish the true performance gap between Panda and Chronos on standard non-chaotic benchmarks across the full range of prediction horizons used in the literature.

3.2.2 Method

Datasets: ETTh1 (7 channels), ETTh2 (7 channels), Weather (21 channels). Horizons: $H \in \{96, 192, 336, 720\}$. $n_{\text{windows}} = 20$. Per-window normalisation. Wilcoxon signed-rank tests. Weather $H = 720$ was not completed due to hardware time constraints.

3.2.3 Observations

Dataset	H	Panda MAE	IQR	Chronos MAE	IQR	Adv (p)
ETTh1	96	0.7269	0.2203	0.6633	0.2388	−0.064 (0.844)
ETTh1	192	0.8185	0.3046	0.7825	0.2742	−0.036 (0.826)
ETTh1	336	0.8571	0.1971	0.9013	0.1859	+0.044 (0.649)
ETTh1	720	0.9921	0.3226	1.0189	0.3845	+0.027 (0.774)
ETTh2	96	0.8736	0.4790	0.9494	0.5583	+0.076 (0.478)
ETTh2	192	0.9697	0.5380	0.9505	0.5730	−0.019 (0.147)
ETTh2	336	0.9255	0.3860	1.1101	0.3831	+0.185 (0.013)*
ETTh2	720	1.1139	0.2194	1.1027	0.3862	−0.011 (0.392)
Weather	96	0.6378	0.1723	0.8115	0.2036	+0.174 (0.000)*
Weather	192	0.7224	0.2385	0.9582	0.2089	+0.236 (0.001)*
Weather	336	0.8481	0.2898	1.0843	0.3507	+0.236 (0.000)*

Table 2: Fixed Experiment 8. * denotes $p < 0.05$ (Wilcoxon, $n = 20$ windows).

[OBS] On ETTh1, no horizon shows statistically significant difference in either direction. All p-values exceed 0.6. The direction of advantage is inconsistent across horizons. **[OBS]** On ETTh2, one horizon ($H = 336$) shows significant Panda advantage ($p = 0.013$). Other horizons are not significant and inconsistent in direction. **[OBS]** On Weather, Panda has statistically significant advantage at $H = 96, 192, 336$ (all $p \leq 0.001$). Advantage magnitude is 0.17–0.24, consistent across horizons. **[PAT]** Weather is the only dataset where Panda shows consistent, statistically significant advantage across multiple horizons.

3.2.4 Competing Explanations for the Weather Result

1. **[HYP]** Weather has 21 channels versus 7 for ETT. More channels provide more information regardless of coupling. Any multivariate model might benefit.
2. **[HYP]** Weather has genuine multivariate physical coupling (pressure drives wind drives temperature). Channel attention may capture this.
3. **[HYP]** Chronos is specifically poorly suited to Weather’s complex multi-scale spectral structure (10-minute sampling, 144-step daily cycle).
4. **[SPEC]** Panda handles quasi-chaotic components of Weather (atmospheric turbulence at synoptic scales) better than Chronos due to its chaotic training distribution.

Confidence: High. for the Weather observation ($n = 20$, $p < 0.001$). **Confidence: Low.** for any proposed mechanism.

3.3 Experiment 9: Univariate Ablation on Weather

3.3.1 Motivation

To directly test whether channel attention drives Panda’s advantage on Weather. If channel attention is responsible, removing it (by forecasting each channel independently) should substantially reduce Panda’s advantage over Chronos.

3.3.2 Method

Multivariate Panda (all 21 channels processed jointly) was compared against univariate Panda (each channel processed independently, suppressing cross-channel attention) at $H \in \{96, 336\}$. $n_{\text{windows}} = 8$.

3.3.3 Observations

H	panda_uni	panda_multi	Adv (uni better)	p
96	0.5541	0.6113	+0.057	0.074
336	0.8467	0.8762	+0.030	0.371

Table 3: Univariate ablation. Positive advantage means univariate Panda has lower MAE than multivariate Panda.

[OBS] At both horizons, univariate Panda has lower MAE than multivariate Panda. **[OBS]** Neither result is statistically significant at $\alpha = 0.05$ ($p = 0.074$, $p = 0.371$). **[OBS]** Effect size is small (0.03–0.06 MAE difference).

3.3.4 What This Establishes and Does Not Establish

The data is inconsistent with the hypothesis that channel attention drives Panda’s Weather advantage. Channel attention does not help and may marginally hurt on Weather. However the effect is small and not significant at $n = 8$ windows. A definitive conclusion requires more windows.

3.3.5 Competing Explanations

1. **[HYP]** The temporal architecture (non-causal encoder, Koopman patch embedding, fixed prediction head) is sufficient to beat Chronos. Channel attention adds noise on non-chaotic data. **Confidence: Medium.:** consistent with data.
2. **[HYP]** $n = 8$ is too small to detect a real but small positive channel attention effect. The true effect may be positive but undetectable. **Confidence: Medium.:** cannot rule out.
3. **[SPEC]** Channel attention trained on chaotic ODE coupling learns patterns counterproductive on periodic non-chaotic data. **Confidence: Low.:** speculative.

Confidence: Medium. for the observation. The channel attention hypothesis is not supported but not conclusively falsified.

3.4 Experiment 10: Burgers Viscosity Sweep (Fixed)

3.4.1 Motivation

To test the chaos-threshold hypothesis for PDEs with a proper non-chaotic baseline. The stable solver prevents divergence at high ν , enabling comparison across the full smooth-to-chaotic range.

3.4.2 Method

$\nu \in \{2.0, 1.0, 0.5, 0.1, 0.05, 0.02, 0.01, 0.005\}$. Adaptive timestep solver (CFL stability condition satisfied at all ν). PCA to 16 channels. $n_{\text{windows}} = 8$. $H = 128$.

3.4.3 Observations

ν	Physical regime	Panda MAE	Chronos MAE	Adv (p)
2.000	Diffusion dominated	0.0152	0.0191	+0.004 (0.273)
1.000	Diffusion dominated	0.0186	0.0568	+0.038 (0.004)*
0.500	Transition	0.0402	0.1023	+0.062 (0.004)*
0.100	Weakly nonlinear	0.1127	0.2239	+0.111 (0.004)*
0.050	Shock forming	0.1374	0.2865	+0.149 (0.004)*
0.020	Strong shocks	0.1394	0.2593	+0.120 (0.004)*
0.010	Strong shocks	0.1375	0.2325	+0.095 (0.004)*
0.005	Strong shocks	0.1509	0.2732	+0.122 (0.004)*

Table 4: Fixed Burgers viscosity sweep. * denotes $p = 0.004$, the minimum achievable with $n = 8$ windows.

[OBS] At $\nu = 2.0$, Panda advantage is +0.004, $p = 0.273$. Not significant. This is the strongly diffusion-dominated regime. **[OBS]** At $\nu = 1.0$, $p = 0.004$. Significant. Burgers at $\nu = 1.0$ is diffusion-dominated and non-chaotic by standard criteria ($\lambda_1 \leq 0$). **[OBS]** Panda wins significantly at $\nu = 1.0$ and $\nu = 0.5$, which are not chaotic regimes. **[OBS]** Advantage is not monotonically increasing with decreasing ν . It peaks near $\nu = 0.05$ (0.149), drops at $\nu = 0.02$ (0.120) and $\nu = 0.01$ (0.095), then rises at $\nu = 0.005$ (0.122).

3.4.4 Critical Finding

[PAT] The chaos-threshold pattern observed in the Lorenz ρ sweep does not replicate cleanly in the Burgers viscosity sweep. Panda wins significantly at non-chaotic viscosity values ($\nu = 1.0$, $\nu = 0.5$) where a chaos-specific mechanism would predict no advantage.

3.4.5 Competing Explanations

1. **[HYP]** Even at $\nu = 1.0$, the PCA channels of the Burgers field are spatially coupled through the diffusion operator. Panda’s channel attention may capture coupling that is present even in non-chaotic smooth flows. **Confidence: Medium..**
2. **[HYP]** Chronos is poorly suited to PCA modal time series regardless of chaoticity. The advantage reflects Chronos weakness rather than Panda strength in the non-chaotic regime. **Confidence: Medium.:** consistent with decomposition results.
3. **[HYP]** The single non-significant result at $\nu = 2.0$ versus the significant result at $\nu = 1.0$ may be noise. The true threshold may lie between 1.0 and 2.0 rather than at the expected chaos onset. **Confidence: Medium.:** cannot rule out with $n = 8$.
4. **[SPEC]** Burgers PCA modes may have spectral characteristics that look dynamically complex to Panda even at non-chaotic ν , due to the mode-mixing nature of PCA on a spatially heterogeneous field.

What would distinguish these: Estimating λ_1 from the PCA component time series at each ν value using the corrected Rosenstein estimator. If $\lambda_1 < 0$ at $\nu = 1.0$ but Panda still wins significantly, explanations 1, 2, or 3 are supported over a chaos-specific mechanism.

Confidence: Medium. for the observation. **Confidence: Low.** for any specific mechanism.

3.5 Experiment 11: FFT Decomposition Preprocessing (Fixed)

3.5.1 Motivation

To test whether removing periodic and trend components from a time series before presenting it to Panda changes Panda’s relative advantage. An earlier version of this experiment (Week 1) contained oracle leakage: the future trend and seasonal components were extracted from the full series and given directly to both models. The fixed version uses only context-window statistics.

3.5.2 Method

FFT decomposition applied to each channel of the context window only. Deterministic component (trend plus seasonal harmonics) projected forward naively: linear trend extrapolation plus repetition of the last full seasonal cycle. Both models forecast the residual; the projected deterministic is added back before evaluation. Comparison against vanilla (no decomposition) run in the same evaluation loop. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

3.5.3 Note on the Invalid Earlier Version

The oracle-leaking version of this experiment reported a 65% reduction in MAE for both models after decomposition. The corrected version shows a 25% increase in Panda MAE on ETTh1 at $H = 336$. The two results are in opposite directions. This confirms that

the earlier result was entirely an artifact of providing future periodic information to both models.

3.5.4 Observations

Dataset	Cond.	H	Panda	Chronos	Adv (p)
ETTh1	Vanilla	96	0.7205	0.8157	+0.095 (0.422)
ETTh1	Decomp	96	0.8032	0.8293	+0.026 (0.578)
ETTh1	Vanilla	336	0.8407	0.8856	+0.045 (0.273)
ETTh1	Decomp	336	1.0474	1.0174	−0.030 (0.770)
ETTh2	Vanilla	96	0.9329	1.0409	+0.108 (0.230)
ETTh2	Decomp	96	0.8856	0.9640	+0.078 (0.273)
Weather	Vanilla	96	0.6082	0.6880	+0.080 (0.020)*
Weather	Decomp	96	0.6771	0.6871	+0.010 (0.230)
Weather	Vanilla	336	0.8762	0.9987	+0.114 (0.004)*
Weather	Decomp	336	1.0609	1.0760	+0.015 (0.020)*

Table 5: Fixed FFT decomposition experiment. * denotes $p < 0.05$.

[OBS] Decomposition increases absolute MAE for both models on ETTh1 at $H = 336$ (Panda +25%, Chronos +15%) and on Weather at $H = 96$ (Panda +11%, Chronos +0.1%). **[OBS]** On Weather, decomposition collapses Panda’s advantage. Vanilla $H = 96$: advantage = +0.080, $p = 0.020$. Decomp $H = 96$: advantage = +0.010, $p = 0.230$. **[OBS]** At $H = 336$ on Weather, the decomp advantage remains statistically significant ($p = 0.020$) but the effect size is very small (+0.015 versus vanilla +0.114).

3.5.5 Competing Explanations

1. **[HYP]** Panda’s advantage on Weather comes from better handling of the periodic and trend components. On the residual, both models are comparable. **Confidence: Medium.**: consistent with all available data.
2. **[HYP]** The naive FFT projection introduces large errors in the projected deterministic component, degrading both models. The result reflects projection error rather than genuine comparison on the residual. **Confidence: Medium.**: the 25% MAE increase on ETTh1 is suspicious and may reflect projection error.
3. **[SPEC]** The periodic component of Weather contains genuine multivariate coupling. Removing it leaves a weakly coupled residual where channel attention is irrelevant.

Confidence: High. for the observation. **Confidence: Low.** for any mechanism.

3.6 Experiment 12: Subsampling Methods (Fixed)

3.6.1 Motivation

To test whether dynamical diversity subsampling of spatial locations gives Panda a larger advantage than uniform or variance-stratified subsampling. The variance-stratified baseline was introduced to control for the confound in the earlier experiment, where a degenerate near-zero-variance location artificially improved both models.

3.6.2 Method

Four methods: (1) Uniform spacing, (2) Variance-stratified uniform (excludes bottom 10% variance locations before uniform spacing), (3) PCA (global spatial modes), (4) Diversity (farthest-point sampling in a dynamical feature space: standard deviation, mean absolute value, 90th percentile amplitude, spectral entropy). $\nu \in \{0.05, 0.005\}$. $N_{\text{channels}} = 16$. $n_{\text{windows}} = 8$. $H = 128$.

3.6.3 Observations

ν	Method	Panda MAE	Chronos MAE	Adv	p
0.05	Uniform	0.0317	0.0765	+0.045	0.004
0.05	Stratified	0.0296	0.0736	+0.044	0.004
0.05	PCA	0.1374	0.2915	+0.154	0.004
0.05	Diversity	0.0317	0.1132	+0.082	0.004
0.005	Uniform	0.0288	0.1696	+0.141	0.004
0.005	Stratified	0.0301	0.1779	+0.148	0.004
0.005	PCA	0.1509	0.3029	+0.152	0.004
0.005	Diversity	0.0300	0.2475	+0.218	0.004

Table 6: Fixed subsampling comparison. All results $p = 0.004$ (minimum achievable with $n = 8$ windows).

[OBS] All results reach $p = 0.004$, the minimum achievable Wilcoxon p-value with $n = 8$ windows. The p-values are therefore uninformative for comparing methods against each other. **[OBS]** Panda absolute MAE across Uniform, Stratified, and Diversity methods is nearly identical at both ν values. At $\nu = 0.005$: Uniform = 0.0288, Stratified = 0.0301, Diversity = 0.0300. **[OBS]** Chronos MAE varies substantially across methods. Diversity locations produce higher Chronos MAE (0.2475) than Uniform locations (0.1696) at $\nu = 0.005$. **[OBS]** Diversity advantage (0.218) exceeds Stratified advantage (0.148) at $\nu = 0.005$. However, the difference is entirely attributable to Chronos MAE (0.2475 vs 0.1779). Panda MAE is the same (0.0300 vs 0.0301).

3.6.4 Critical Observation

[PAT] Panda’s absolute forecasting performance is invariant to the spatial subsampling method (Uniform, Stratified, Diversity). Diversity subsampling increases Panda’s relative advantage over Chronos by making the problem harder for Chronos, not easier for Panda.

3.6.5 Competing Explanations for the Diversity Effect

1. **[HYP]** Diversity locations are more dynamically informative. Chronos cannot handle them but Panda can, demonstrating a genuine advantage of Panda’s architecture. **Confidence: Low.:** speculative, not supported by Panda’s flat absolute MAE.
2. **[HYP]** Diversity sampling selects spatially extreme or high-variance locations that break Chronos pattern matching without providing additional information to Panda. **Confidence: Medium.:** consistent with the observation that Panda MAE is unchanged.

3. **[HYP]** PCA gives the largest relative advantage because global modes are harder for Chronos (high complexity, global structure) while Panda handles them adequately due to the Koopman embedding. **Confidence: Medium..**

Confidence: Medium. for the invariance of Panda absolute MAE observation. **Confidence: Low.** for any proposed mechanism.

4 Summary of Findings

4.1 Solid Observations (High Confidence)

1. Panda has statistically significant MAE advantage over Chronos on Weather at $H = 96, 192, 336$ ($n = 20, p \leq 0.001$). Advantage magnitude 0.17–0.24.
2. Channel attention does not drive this advantage. Univariate Panda performs comparably to or marginally better than multivariate Panda on Weather at $H = 96$ and $H = 336$ ($n = 8$, non-significant).
3. Removing periodic components from Weather via FFT nearly eliminates Panda’s advantage over Chronos (vanilla +0.080 versus decomp +0.010 at $H = 96$).
4. Panda wins on all tested chaotic **dysts** and Lorenz systems ($n = 8$ per system, no counter-example found).
5. The Lorenz ρ sweep shows a pattern consistent with a threshold near $\rho = 24.74$.

4.2 Medium Confidence Observations

1. Panda wins on Burgers at $\nu = 1.0$ (non-chaotic), suggesting a chaos-specific mechanism does not fully explain the PDE advantage.
2. Panda’s advantage on Burgers does not increase monotonically with decreasing viscosity.
3. Diversity subsampling increases Panda relative advantage over Chronos by making the problem harder for Chronos, not by helping Panda.
4. Panda absolute MAE is invariant to spatial subsampling method (Uniform, Stratified, Diversity).

4.3 Not Established Despite Earlier Discussion

1. The mechanism behind Panda’s Weather advantage.
2. Whether chaos is necessary and sufficient for Panda’s advantage (Burgers result partially contradicts this for PDEs; Weather result is ambiguous).
3. Whether channel attention is the mechanism on any dataset.
4. Whether the Lorenz bifurcation result is chaos-specific or signal-statistics-specific.
5. Whether the Spearman correlation between λ_1 and Panda advantage is real (p-hacking issue).

5 Proposed Next Experiments

The following experiments are proposed in priority order, based on the open scientific questions identified above.

5.1 Priority 1: Understand the Weather Mechanism

The most important unresolved question is what drives Panda’s consistent, significant advantage on Weather when channel attention is not the driver and removing periodicity nearly eliminates the advantage.

Proposed experiment: Direct periodic component forecasting. Run both models on Weather using only the FFT-extracted deterministic component (trend plus seasonal harmonics) as both input and target. Compare Panda advantage on the deterministic component versus the residual. If Panda advantage is large on the deterministic part and near zero on the residual, it establishes that Panda’s advantage comes from handling periodic structure, not dynamical residuals. This directly distinguishes between explanations 1 and 2 from Experiment 11.

5.2 Priority 2: Rule Out Signal Statistics in Lorenz Sweep

Proposed experiment: Signal-statistics-matched Lorenz control. At each ρ value, rescale the normalised Lorenz trajectory to have identical variance and spectral entropy to the $\rho = 15$ (periodic) baseline trajectory before presenting to both models. If the advantage pattern persists after matching, signal statistics are not the driver. If it disappears, signal statistics were confounding the result.

This experiment requires no new simulation or model loading, only preprocessing of existing trajectories.

5.3 Priority 3: Verify the Burgers Non-Chaotic Result

Proposed experiment: Estimate λ_1 from saved Burgers PCA component time series at each ν value using the corrected Rosenstein estimator. If $\lambda_1 < 0$ at $\nu = 1.0$ and $\nu = 0.5$ but Panda still wins significantly, the chaos-specific mechanism is directly falsified for PDEs.

5.4 Priority 4: Evaluate Diversity Subsampling Reliability

Proposed experiment: Run Diversity and Stratified Uniform subsampling at 10 different random seeds at $\nu = 0.005$. Measure variance in Panda absolute MAE and Panda advantage across seeds. This determines whether the single-seed result is reliable and whether diversity subsampling genuinely provides advantages beyond avoiding degenerate locations.

5.5 Priority 5: Improved Period Projection

Proposed experiment: Rerun Experiment 11 with a better projection method for the deterministic component: instead of naive seasonal repetition, use the actual seasonal pattern from the full series up to the context window end (not oracle leakage since no future residual values are used). If Panda’s advantage survives this improved projection,

the residual contains genuine predictable structure that Panda handles better. If the advantage collapses again, Panda’s advantage is genuinely in the periodic components. This result directly determines whether TimesNet-inspired pretraining (which teaches models about periodic structure) would benefit Panda.